



Vardan Pal

Software Engineering Intern | Full Stack Developer |
Backend Engineer
B.Tech – Information Technology
2022 – 2026
National Institute of Technology, Srinagar

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PROFESSIONAL SUMMARY

Final-year B.Tech (IT) student at NIT Srinagar with professional software engineering internship experience delivering cloud-native backend systems in production. Shipped 20+ RESTful API endpoints across 3 full-stack projects, integrated AI/OCR pipelines achieving ~98% accuracy, and containerized services using Docker. Proficient in MERN stack, secure API design, JWT-based authentication & authorization, and scalable system architecture. Passionate about engineering production-ready software that solves real problems.

EDUCATION

- **National Institute of Technology, Srinagar** 2022 – 2026
Bachelor of Technology – Information Technology
– Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, System Design
- **Army Public School** 2022
Class XII – CBSE | 87.4%
- **Army Public School** 2020
Class X – CBSE | 91.4%

EXPERIENCE

- **Noventiq (Umbrella Infotech)** Dec 2024 – Feb 2025
Software Engineering Intern – Cloud Native Applications Noida, India
– Engineered 8+ modular RESTful API endpoints for enterprise cloud-native applications using Node.js and Express.js; optimized middleware execution pipeline reducing average response latency by eliminating redundant query patterns.
– Containerized 3 backend microservices using Docker, standardizing deployment across development and staging environments and eliminating environment-specific runtime failures entirely.
– Implemented secure coding practices – input validation, parameterized queries, and token-based authentication & authorization flows – across 5+ service modules, strengthening the application's overall security posture.
– Collaborated in Agile/Scrum sprint cycles: conducted peer code reviews, refactored 2 legacy modules for improved maintainability, and resolved critical production-blocking defects under tight delivery timelines.
– Optimized MongoDB query performance by restructuring collection schemas and introducing compound indexing strategies, improving data retrieval efficiency by a measurable margin for high-frequency API endpoints.

PROJECTS

- **HealthEase – AI-Powered Telemedicine Platform** Jan 2025 – Jun 2025
React, Node.js, Express.js, MongoDB, Docker, GPT-4o, Cloud Vision API, JWT, RBAC github.com/Vardxn/HealthEase
– Architected a multi-role telemedicine platform with service-oriented backend design; implemented separate RESTful API layers for Patient, Doctor, and Admin roles with JWT authentication and RBAC middleware enforcing access control on every route.
– Designed and integrated an OCR pipeline using Google Cloud Vision API to digitize handwritten medical prescriptions, achieving ~98% extraction accuracy across structured and semi-structured prescription formats.
– Engineered an AI-powered medical chatbot by integrating OpenAI GPT-4o with a custom prompt engineering layer, enabling context-aware patient queries with domain-specific guardrails for safe information delivery.
– Delivered a secure REST API backend with 15+ endpoints covering appointments, prescriptions, medication reminders, and user management; applied input sanitization and role-gated route protection on all layers.
– Containerized the full application stack using Docker with environment-specific .env configurations, enabling one-command reproducible local setup and consistent CI/CD-ready deployment.
- **Wanderlust – Full-Stack Travel Listing Platform** Aug 2023 – Dec 2023
Node.js, Express.js, MongoDB, Passport.js, EJS, Cloudinary, Mapbox API github.com/Vardxn/Wanderlust
– Designed and implemented a production-grade MVC-architected backend with clearly separated route handlers, middleware chains, and Mongoose model layers; achieved a maintainable, testable codebase structure scalable to additional feature modules.
– Engineered session-based authentication & authorization using Passport.js (local strategy) with bcrypt password hashing, persistent login sessions, and protected route middleware across all user-facing endpoints.
– Integrated Mapbox Geocoding API for geolocation-based listing discovery and Cloudinary CDN for optimized media storage and delivery, reducing image load latency through cloud-based asset transformation pipelines.

- Implemented complete CRUD operations for 50+ travel listings with server-side validation, error-handling middleware, and flash messaging, supporting multi-category browsing, user reviews, and booking workflows.

•Mohishree Facility Services – B2B Industrial Marketplace

Jun 2024 – Sep 2024

Next.js, TypeScript, Node.js, PostgreSQL, Tailwind CSS, REST APIs

github.com/Vardxn/Mohishree

- Architected a B2B marketplace connecting industrial clients with skilled service providers; designed a normalized PostgreSQL relational schema with role-based access control for Client, Vendor, and Admin entities.
- Developed a fully type-safe REST API layer in TypeScript with strict request/response interface contracts, enabling zero runtime type errors and reliable integration between Next.js SSR frontend and Node.js backend.
- Implemented server-side rendering (SSR) with Next.js for SEO-critical service category pages; applied dynamic routing and static generation strategies to reduce time-to-interactive on first load by eliminating client-side data fetching bottlenecks.
- Engineered multi-step vendor onboarding workflows with form validation, document upload handling via Multer, and admin approval pipelines – supporting end-to-end B2B service procurement flows from registration to contract.

TECHNICAL SKILLS

Languages: C++, JavaScript (ES6+), TypeScript, Python

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3, EJS

Backend: Node.js, Express.js, RESTful API Design, JWT Authentication & Authorization, RBAC, Middleware Architecture, Microservices

Databases: MongoDB, PostgreSQL, Mongoose ODM, Database Indexing, Schema Design

Cloud & DevOps: Docker, Git, GitHub, CI/CD Pipelines, Linux, AWS EC2, AWS S3, Postman

AI & APIs: OpenAI API (GPT-4o), Google Cloud Vision API, OCR Integration, Prompt Engineering

Developer Tools: VS Code, GitHub Copilot, npm, Cloudinary, Mapbox API, Multer

Concepts: System Design, Scalability, Secure Coding, MVC Architecture, Version Control, Agile/Scrum

ACHIEVEMENTS & RECOGNITION

•Software Engineering Intern– Noventiq (Umbrella Infotech), Noida

Dec 2024 – Feb 2025

- Competitively selected for a professional SWE internship; contributed to enterprise cloud-native backend systems, REST APIs, and Docker-based deployments in a cross-functional Agile team environment.

•3 Production-Grade Projects Shipped– Full Stack, AI, and B2B Domains

2023 – 2025

- Independently architected and delivered 3 end-to-end applications spanning telemedicine AI, travel marketplace, and industrial B2B – collectively covering 20+ REST endpoints, OCR pipelines, and multi-role RBAC systems.

•5+ Public Repositories on GitHub– github.com/Vardxn

2023 – Present

- Maintained consistent open-source presence with documented repositories across MERN, Next.js, and TypeScript stacks; projects demonstrate end-to-end software engineering from architecture to deployment.

CERTIFICATIONS

•Full Stack Web Development – Sigma 8.0– Apna College By Mr. Aman Dhatarwal(Completed)

2024

- Comprehensive MERN stack curriculum covering DSA, system design, backend engineering, REST APIs, and full deployment workflows.

•AI/ML Prime– Apna College By Mr. Aman Dhatarwal(In Progress)

2024 – 2025

- Machine learning fundamentals, Python data science libraries (NumPy, Pandas, Matplotlib), and AI-powered application development.

•Software Engineering Job Simulation– JPMorgan Chase & Co. via Forage (Completed)

Jan 2026

- Completed enterprise-grade software engineering tasks covering Kafka integration, H2 in-memory database setup, REST API design, and REST API controller implementation in a simulated JPMorgan production environment.